

Rashmi Shaw

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leetcode.com/u/rinki_2005

Professional Profile

Final-year IT student with hands-on experience building AI-powered and data-driven applications. Proficient in Python and deep learning frameworks (PyTorch, TensorFlow), with practical exposure to NLP, LLM integration, and computer vision pipelines. Passionate about predictive modeling, feature engineering, and deriving actionable insights from data.

Education

National Institute of Technology, Kurukshetra
B.Tech in Information Technology (CGPA: 8.0/10)

2022 - 2026
India

Technical Skills

Languages & Data Tools: Python, SQL, C++, Java, JavaScript (ES6+) | Pandas, NumPy, Matplotlib, Seaborn

ML / AI / NLP: PyTorch, TensorFlow, OpenCV, YOLO | LLM integration (Ollama), Generative AI, NLP pipelines, feature engineering, predictive modeling

Backend & Infra: Node.js, Express.js, REST APIs, Kafka, Redis, PostgreSQL, MongoDB, Docker, Git

Frontend: React.js, Next.js, Tailwind CSS **DSA:** Graphs, DP, Trees, Heaps, Tries, Greedy, Sliding Window

Projects

Basketball Video Analysis - Computer Vision & Predictive Modeling 2024

- Built an end-to-end ML pipeline using **YOLO and OpenCV**, achieving **95% player detection accuracy** across real-match footage.
- Performed **feature engineering** on player trajectory data for multi-object tracking, team classification, and ball possession prediction.
- Analyzed detection confidence distributions and tracking **metrics** to iteratively evaluate and improve model performance.

Vibecode Playground - AI-Powered Online Coding Platform 2025

- Built a full-stack platform (Next.js, React, PostgreSQL) with browser-based code execution and multi-user project management.
- Integrated a local **LLM assistant (Ollama)** for **conversational AI** support inside the IDE; experimented with **AI automation workflows** and built usage **dashboards and actionable reports** to surface platform insights.
- Designed REST APIs with OAuth login and versioned file storage; cut developer onboarding from **30 minutes to under 1 minute**.

JobTaker - Distributed Job Portal (Microservices) 2025

- Designed a **microservices-based backend** (Node.js, PostgreSQL, Kafka, Redis) with event-driven architecture for independent service scaling.
- Optimized PostgreSQL queries via indexing and connection pooling, reducing API latency by **65%**; load-tested with k6 at **5,000+ concurrent users** with p95 under **200ms**.
- Implemented **JWT-based auth**, Redis-backed OTP, RBAC, retry mechanisms, and graceful degradation for production reliability.

Problem Solving & Algorithms

Solved **500+ DSA problems** on LeetCode with strong emphasis on Graphs, Dynamic Programming, Trees, Heaps, Tries, Greedy, and Sliding Window. Proficient in BFS, DFS, shortest paths, union-find, topological sorting, and optimization DP patterns.

Leadership & Activities

Core Member - Shiksha (NGO): Led education initiatives impacting **180+ underprivileged students**; coordinated **150+ volunteers** managing scheduling and execution of teaching programs.